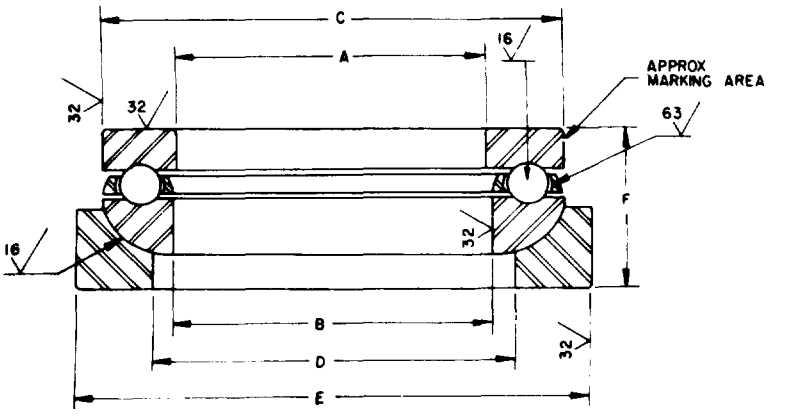
FED SUP CLASS
3110



NOTES:

1. MATERIAL: STEEL, ALLOY, HIGH-CARBON CHROMIUM, TYPE AISI E51100 OR TYPE AISI E52100 OF FEDERAL STD. NO. 66.
2. SURFACE FINISH: SURFACES MARKED ✓ SHALL HAVE SURFACE FINISH IN ACCORDANCE WITH USAS B46.1.
3. HEAT-TREAT: BEARING SHALL BE THROUGH HARDENED TO ROCKWELL C-58 TO C-66.
4. BEARING SHALL BE FREE OF ALL DEFECTS WHICH WOULD AFFECT ITS SERVICEABILITY.
5. ✳ BEARING MUST CLEAR MAXIMUM HOUSING OR SHAFT FILLET RADIUS SHOWN IN FILLET RADIUS COLUMN.
6. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
7. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
8. THE MS PART NUMBER CONSISTS OF THE MS NUMBER, PLUS THE DASH NUMBER. EXAMPLE: MS17102-1.
9. MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.

BEARING LOAD FORMULA:

THE DYNAMIC LOAD RATING (IN POUNDS) FOR ANY COMBINATION OF ROTATIONAL SPEED (IN RPM) AND RATING LIFE (IN HOURS) MAY BE FOUND FROM THE FOLLOWING FORMULA:

$$P_A = \frac{100\ C}{(60\ N\ H)^{1/3}}$$

WHERE:

P_A = REQUIRED DYNAMIC LOAD RATING - POUNDS
C = BASIC DYNAMIC LOAD RATING - POUNDS
N = ROTATIONAL SPEED - RPM
H = RATING LIFE - HOURS

THE ABOVE FORMULA IS IN AGREEMENT WITH THE LATEST AFBMA STANDARD METHOD OF EVALUATING LOAD RATINGS OF THRUST BALL BEARINGS.

BASIC LOAD RATING HAVE ALSO BEEN DETERMINED ACCORDING TO AFBMA STANDARD METHODS.

THE DEFINITION OF RATING LIFE IS THE NUMBER OF HOURS AT SOME CONSTANT SPEED OF THE INNER RING THAT 90 PERCENT OF A GROUP OF APPARENTLY IDENTICAL BEARINGS WILL COMPLETE OR EXCEED BEFORE THE FIRST EVIDENCE OF FATIGUE DEVELOPS.

Ⓐ ENTIRE STANDARD REVISED

APPROVED 11 SEPTEMBER 1959 REVISED Ⓐ 20 FEBRUARY 1968

REVIEW ACTIVITIES: ARMY- WC
USER ACTIVIES: ARMY-MI ; NAVY-YD , MC

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. ARMY-AT Other Cust AF -11	TITLE: BEARING, BALL, THRUST: METRIC, SELF-ALIGNING	MILITARY STANDARD MS17102
PROCUREMENT SPECIFICATION NONE	SUPERSEDES	SHEET 1 OF 2

DD FORM (SEP 67) 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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DD FORM SEP 67 672-1 (Coordinated)

PREVIOUS EDITION OF THIS FORM ARE OBSOLETE

P.A. ARMY-AT Other Cust AF -11

PROCUREMENT SPECIFICATION NONE

SUPERSEDES

TITLE: BEARING, BALL, THRUST, METRIC, SELF-ALIGNING

MILITARY STANDARD

MS17102

SHEET 2

DASH NUMBER	A (NOM) SMALL BORE MM	A (NOM) SMALL BORE INCH	A TOL ±	B LARGE BORE MM	B LARGE BORE +.005 -.000 INCH	C BEARING O.D. MM	C BEARING O.D. INCH	C TOL +.000 -	D WASHER I.D. MM	D WASHER I.D. INCH	E WASHER O.D. MM	E WASHER O.D. +.000 -.002 INCH	F HEIGHT OVERALL MM	F HEIGHT OVERALL INCH	F TOL ±	R FIL. RAD. * MAX	BASIC DYNAMIC LOAD RATING LBS	BASIC STATIC LOAD RATING LBS	MIL-STD-102 CODE NUMBER
1	15	.5906	.0004	17	.669	35	1.3780	.002	22	.866	38	1.4961	17	.669	.005	.02	3610	5620	445x01517x4000
2	20	.7874	.0004	22	.866	40	1.5748	.002	25	.984	45	1.7717	18	.709	.005	.02	3870	6740	445x02056x4000
3	25	.9843	.0004	27	1.063	48	1.8898	.002	29	1.417	50	1.9685	19	.748	.005	.02	4990	9710	445x02586x4000
4	30	1.1811	.0004	32	1.260	53	2.0866	.002	42.5	1.673	59	2.3228	20	.787	.005	.02	5030	10400	445x03026x4000
5	35	1.3780	.0005	37	1.457	62	2.4409	.002	46	1.811	67	2.6378	23	.906	.005	.02	7080	15000	445x03546x4000
6	40	1.5748	.0005	42	1.654	64	2.5197	.002	48	1.890	69	2.7165	23	.906	.005	.02	6280	14300	445x04056x4000
7	45	1.7717	.0005	47	1.850	73	2.8740	.002	53	2.087	78	3.0709	27	1.063	.005	.02	8490	19900	445x04546x4000
8	50	1.9685	.0005	52	2.047	78	3.0709	.002	58	2.283	83	3.2677	27	1.063	.005	.02	9790	23100	445x05036x4000
9	55	2.1654	.0006	57	2.244	88	3.4646	.002	71	2.795	94	3.7008	30	1.181	.005	.02	12000	28400	445x05556x4000
10	60	2.3622	.0006	62	2.441	90	3.5433	.002	69	2.717	96	3.7795	30	1.181	.005	.02	10400	27200	445x06086x4000
11	65	2.5591	.0006	67	2.638	100	3.9370	.002	74	2.913	105	4.1339	34	1.339	.005	.04	12900	33700	445x06526x4000
12	70	2.7559	.0006	72	2.835	103	4.0551	.002	81	3.189	109	4.2913	34	1.339	.005	.04	13200	35500	445x07056x4000
13	75	2.9528	.0006	77	3.031	110	4.3307	.002	89	3.504	114	4.4882	34	1.339	.005	.04	13300	37300	445x07526x4000
14	80	3.1496	.0006	82	3.228	115	4.5276	.002	89	3.504	124	4.8819	37	1.457	.005	.04	15400	44100	445x08036x4000
15	85	3.3465	.0008	88	3.465	125	4.9213	.002	98	3.858	138	5.4331	40	1.575	.010	.04	19200	52700	445x08546x4000
16	90	3.5433	.0008	93	3.661	135	5.3150	.002	110	4.331	141	5.5512	40	1.575	.010	.04	23000	63800	445x09066x4000
17	95	3.7402	.0008	98	3.858	140	5.5118	.002	107	4.213	151	5.9449	43.5	1.713	.010	.04	24900	69700	445x09586x4000
18	100	3.9370	.0008	103	4.055	150	5.9055	.003	118	4.646	156	6.1417	44	1.732	.010	.04	27600	79900	445x10046x4000
19	105	4.1339	.0008	108	4.252	155	6.1024	.003	118	4.646	163	6.4173	49	1.929	.010	.04	31300	89100	445x10576x4000
20	115	4.5276	.0008	118	4.646	165	6.4961	.003	128	5.039	173	6.8110	52	2.047	.010	.04	31700	93700	445x11556x4000
21	125	4.9213	.0010	128	5.039	175	6.8898	.003	139	5.472	186	7.3228	56	2.205	.010	.04	33100	103000	445x12576x4000
22	140	5.5118	.0010	143	5.630	200	7.8740	.003	156	6.142	212	8.3464	62	2.441	.010	.04	41800	131000	445x14016x4000

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APPROVED 11 SEPTEMBER 1959 REVISED (A) ENTIRE STANDARD REVISED

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